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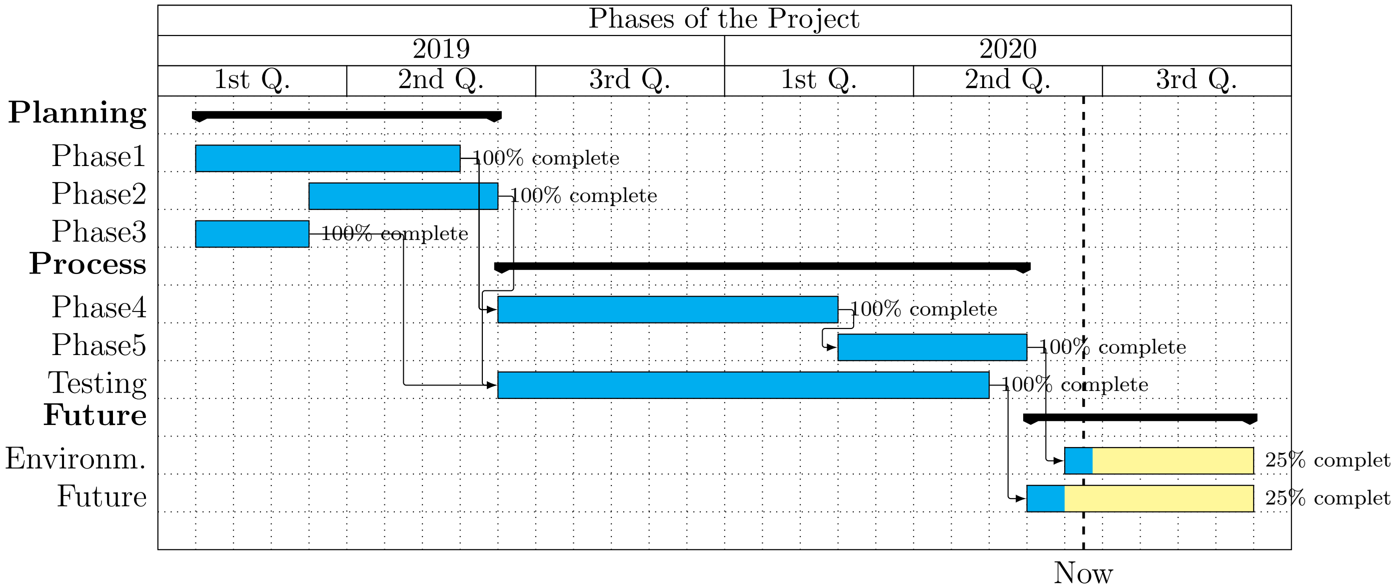

Figure 1: Gantt diagram of the project

For more information read the manual [1] of Skala.

2 State of the Art

This chapter builds the scientific and technical foundation for the work described later in this thesis. The central question is deceptively simple: *can a short voice recording tell a clinician something meaningful about what is happening inside a patient's brain?* Answering it requires three kinds of understanding: the disease biology that shapes how patients speak, the track record of computational approaches that have tried to extract signal from that speech, and the engineering toolkit that modern deep learning makes available. The chapter addresses each in turn, following the structure of recent surveys [16] while staying anchored to the specific threads of this thesis [5]: pathology-aware biomarker estimation, differential diagnosis of primary progressive aphasia (PPA), cross-lingual dataset gaps, self-supervised encoders, multimodal fusion, and handcrafted paralinguistic descriptors.

2.1 Clinical and Biological Background

Before asking how a machine should analyse speech, it is worth asking what the speech is actually reflecting. Neurodegenerative diseases do not affect language randomly—they follow anatomical and proteinopathic patterns that produce characteristic breakdowns in fluency, grammar, and lexical access. Understanding those patterns is what makes the computational problem tractable rather than arbitrary.

2.1.1 Amyloid pathology and Alzheimer's disease

Alzheimer's disease kills neurons silently for years before any symptom appears. By the time a patient first complains of memory problems, cortical atrophy may already be widespread and therapeutic windows may have narrowed. The underlying molecular event—abnormal aggregation of amyloid-β (Aβ) peptides—was identified three decades ago [6], yet the path from that discovery to early, scalable detection has remained frustratingly slow.

Today, confirmatory biomarker assessment relies on cerebrospinal fluid (CSF) assays or positron emission tomography (PET) imaging. Both are accurate. Both are also expensive, capacity-constrained, or invasive—lumbar puncture to collect CSF is a procedure many patients decline, while PET scanners are rare outside major centres. The result is that large fractions of at-risk populations reach specialist clinics only after substantial decline has already occurred. This access gap is the primary clinical motivation for exploring speech as a low-cost screening signal.

[Figure: Biomarker modalities plotted by invasiveness and cost—speech sits at the low-burden extreme]

Figure 2: Speech-based biomarkers occupy a unique position in the cost–invasiveness landscape, enabling frequent and remote monitoring that PET or CSF cannot match.

2.1.2 Primary progressive aphasia and its variants

A parallel diagnostic challenge arises at the syndrome level. Language deficits emerge across neurodegenerative aetiologies with overlapping phenotypes, making early differential diagnosis difficult even for experienced neurologists. Primary progressive aphasia (PPA) exemplifies this tension: it is a disorder in which progressive language impairment is the dominant complaint, and it presents in three broadly recognised variants [7, 8].

The non-fluent/agrammatic variant (*nfv*PPA) is characterised by effortful, halting speech and grammatical breakdown. Patients produce short, telegraphic utterances; verb morphology and function words are the first casualties. The semantic variant (*sv*PPA) spares speech rhythm entirely but erodes word and concept meaning—patients speak fluently but substitute or lose specific nouns, and their knowledge of object categories deteriorates. The logopenic variant (*lv*PPA) sits between the two: speech rate is reduced and frequent mid-sentence pauses interrupt otherwise grammatical output, while repetition and phonological processing are selectively impaired.

These profiles are not merely academic labels. They carry probabilistic information about the underlying proteinopathy—*nfv*PPA associates with tau inclusions; *sv*PPA with TDP-43 pathology; *lv*PPA predominantly with amyloid-β [9]—which in turn informs prognosis and, increasingly, treatment eligibility. Importantly, variant boundaries can blur early in the disease course, making quantitative assessment particularly valuable [26].

[Figure: Clinical and linguistic profiles of the three PPA variants]

Figure 3: Comparison of linguistic breakdown patterns, associated proteinopathies, and neuroanatomical correlates in the three PPA variants [7, 9].

From an engineering perspective, these clinical facts carry a direct implication: a useful diagnostic system must be sensitive to *how* a person speaks (articulatory tempo, pause placement, grammatical structure) *and* to *what* they say (lexical choice, semantic coherence, information density). Neither the acoustic nor the linguistic channel alone tells the full story [10].

2.2 Speech as a Digital Biomarker

Voice has a practical advantage that no blood draw or imaging scan can match: it is already there. Patients speak during every clinical visit, and recordings can be collected remotely, repeatedly, and at negligible marginal cost. This section traces the arc of computational research that has tried to realise that promise—from early hand-crafted feature pipelines, through community benchmarks that enabled reproducible comparison, to the most recent data-driven classifiers for PPA variant differentiation.

2.2.1 From engineered features to deep representations

Early computational work focused on features extractable from transcripts: lexical diversity indices, syntactic complexity measures, and idea-density scores derived from parse trees [12]. Shallow classifiers—support-vector machines, random forests—trained on these representations demonstrated that linguistic signal exists in spontaneous speech and could separate AD patients from healthy controls at above-chance accuracy. The approach was important as a proof of concept.

Its weakness was practical. Manual transcription is slow and expensive. The engineered feature sets required considerable domain expertise to define and maintain, and their coverage of the acoustic signal was fragmentary—the temporal dynamics of the raw waveform went largely unused. The resulting decision boundaries were shallow, and performance plateaued as more data or richer feature engineering yielded diminishing returns [10, 13, 14, 15].

The natural response was to let models learn their own representations. Rather than specifying which acoustic or linguistic properties matter, representation learning discovers structure from data—either end-to-end on labelled examples or, more powerfully, through self-supervised pre-training on large unlabelled corpora. Reviews of speech-based AD detection now consistently report two findings [16]: deep neural pipelines dominate leaderboard benchmarks when adequate labelled data exist, and systems that jointly consume acoustic and text modalities outperform either channel alone. These trends directly motivate the architectures explored in this thesis [5].

2.2.2 Benchmark challenges and public corpora

Progress in any empirical field depends on shared benchmarks. Without frozen evaluation splits and a common protocol, it is impossible to tell whether a new system genuinely improves over the state of the art or merely exploits a more favourable data partition. This was the motivation behind the ADReSSo challenge [17], which drew a speech-only subset from the DementiaBank Pitt corpus—picture-description recordings balanced by age and gender—and defined a public leaderboard for AD versus healthy-control classification. Competition entries established strong baselines [27], and subsequent work continues to probe model interpretability and feature attribution on the same frozen split [28].

ADReSSo has been valuable, but it also has well-known limitations. It is English only. It captures a single elicitation task—the Cookie Theft picture description. Its size, roughly 156 speakers, limits the diversity of patient profiles that can be represented. Performance on this corpus should therefore be interpreted carefully: it reflects generalisation within a particular recording condition and demographic, not deployment in an arbitrary clinic.

Initiatives such as the Sant Pau Initiative on Neurodegeneration (SPIN) [18] address this gap by assembling multicentre longitudinal data spanning genetics, imaging, fluid markers, and structured cognitive assessments in predominantly Catalan- and Spanish-speaking populations. Subsets incorporating spontaneous speech tied to standard batteries—picture-description prompts compatible with the Western Aphasia Battery [29]—provide the language-specific evaluation ground that English-only benchmarks cannot offer. Spanish-focused studies confirm that language-aware descriptors achieve strong variant discrimination in PPA cohorts [30], while bilingual analyses add a cautionary note: acoustic models may retain utility when patients speak a non-dominant language, but lexical models become brittle—a critical consideration for the multilingual clinical realities of Catalonia [31].

2.2.3 Automatic PPA variant classification

Identifying that a patient has a neurodegenerative language disorder is only the first step. Knowing *which* variant matters clinically, because variants carry different prognoses and—with disease-modifying therapies entering trials—different treatment pathways. This creates a multi-class classification problem with an inherently noisy ground truth: clinical labels are assigned by consensus after detailed neuropsychological testing, yet boundaries between variants blur, particularly early in disease course [26].

Empirical studies map specific speech features onto the neurolinguistic breakdown profiles hypothesised for each variant. Reduced speech rate and grammatical errors mark *nfv*PPA. Semantic word substitutions and empty filler words characterise *sv*PPA. Frequent within-sentence pauses, phonological errors on multi-syllabic words, and impaired sentence repetition point toward *lv*PPA. Quantitatively, pause structure, filler usage, lexical diversity, and discourse coherence metrics are discriminative at the group level [32], while raw articulation tempo alone separates non-fluent from fluent variants with clinically relevant accuracy [33, 34]. Connected speech features also associate with cortical amyloid burden within PPA cohorts [15], suggesting that speech encodes not only syndrome identity but correlates of the underlying molecular pathology.

More recent approaches escalate modelling complexity. Large language models combined with curated linguistic measurements achieve high retrospective classification accuracy [35]. Neural architectures trained jointly on acoustic and linguistic embeddings approach expert-level multi-class accuracy in prospective evaluations [36]. Nevertheless, models trained on small cohorts remain sensitive to sample size and label noise, and cross-cohort validation across recording conditions and accents remains a largely open engineering problem [37].

2.3 Deep Learning Architectures for Clinical Speech

The shift toward deep representation learning introduced a new set of modelling choices: which pre-trained encoder to use for audio, which for text, how to fuse the two streams, and whether to retain interpretable handcrafted features alongside opaque neural embeddings. This section covers each choice in turn. Figure 4 provides an overview of how the components fit together.

[Figure: End-to-end multimodal pipeline—from raw audio to clinical predictions]

Figure 4: End-to-end architecture for multimodal clinical speech analysis. Raw audio feeds a self-supervised acoustic encoder; after automatic speech recognition, the transcript feeds a pre-trained text encoder. Both streams are fused and passed to task-specific prediction heads.

2.3.1 Self-supervised acoustic encoders

Self-supervised learning sidesteps the label scarcity problem by defining a pre-training objective that does not require human annotation. The model is trained to predict or reconstruct part of its own input—for example, predicting masked segments from their unmasked context. Trained on large unlabelled audio corpora (thousands of hours of speech), the encoder learns frame-level representations that generalise to downstream tasks with far smaller labelled datasets. This property is especially valuable in clinical settings, where annotated cohorts are almost always small.

wav2vec 2.0. Baevski et al. [19] introduced a two-stage architecture. A convolutional feature extractor maps raw waveform samples to a sequence of latent representations at approximately 20 ms resolution. These are quantised into a discrete codebook via product quantisation, and the resulting tokens become pseudo-labels. A transformer encoder then processes the latent sequence with random masking applied: the pre-training objective is a contrastive loss that requires the model to identify the correct quantised token for each masked position among a set of distractors. The trained encoder produces contextualised frame embeddings that reflect not only local spectral content but temporal context spanning hundreds of milliseconds.

HuBERT. Hsu et al. [20] replaced the online codebook with offline pseudo-labels obtained by k-means clustering of acoustic features from an earlier iteration of the model itself. This decouples label quality from the representation quality during training and enables iterative refinement: cluster, train, re-cluster. The resulting encoder produces representations that align more closely with phonetic units and have shown strong transfer on speaker-independent downstream tasks.

WavLM. Chen et al. [21] extended the HuBERT framework with a denoising objective: some training utterances are corrupted by additive noise or overlapping speech before masking, and the model must predict the clean token despite the distortion. This regularisation yields representations more robust to recording-condition mismatch—a relevant advantage for clinical audio that may contain ambient noise, microphone variation, or co-occurring voices. WavLM consistently achieves top performance across the SUPERB benchmark suite and has emerged as the preferred backbone in recent clinical speech work.

2.3.2 Contextual text encoders

On the text side, the dominant pre-training paradigm is masked language modelling (MLM). A transformer encoder processes a token sequence with a fraction of tokens randomly replaced by a mask symbol; the pre-training objective is to predict the original token from its bidirectional context. Unlike the left-to-right language models used in generation, bidirectional encoders simultaneously attend to preceding and following tokens, making them well suited to sentence-level classification tasks.

BERT. Devlin et al. [22] demonstrated that large-scale MLM pre-training on general text corpora (BooksCorpus and English Wikipedia) yields contextual token embeddings that fine-tune effectively across a wide range of NLP benchmarks with minimal architectural modification. Applied to clinical speech transcripts, BERT embeddings capture lexical choices, local syntactic patterns, and semantic anomalies—all relevant to cognitive assessment.

DistilBERT. Sanh et al. [23] showed that a six-layer student trained to mimic the output distributions of a twelve-layer BERT teacher retains 97% of its downstream task performance at 40% fewer parameters and 60% faster inference. The reduced footprint makes DistilBERT attractive for systems that must be deployed at scale or on resource-constrained hardware.

ModernBERT. Warner et al. [24] revisited the encoder-only architecture with a range of improvements accumulated since 2019: rotary positional embeddings that extend effective context length well beyond 512 tokens, alternating local and global attention to reduce quadratic complexity, and training on substantially larger and more diverse corpora. For clinical speech, the extended context window is directly relevant: spontaneous picture-description narratives, though short by document standards, benefit from a model that can attend globally across an entire utterance without truncation artefacts.

2.3.3 Multimodal fusion strategies

Fusing acoustic and textual streams is not straightforward. The two modalities operate at fundamentally different resolutions: acoustic frames are typically spaced at 20 ms, while word-level transcripts emerge from an ASR system at intervals of hundreds of milliseconds or more. Before any learning can happen, the system must decide how to reconcile this temporal mismatch.

Three broad fusion philosophies have been explored [16, 5]. **Early fusion** aggregates both streams into a single representation before the classification head, typically by temporal pooling followed by concatenation. It is simple, but pooling discards the sequential structure that may carry diagnostic signal. **Late fusion** maintains separate modality-specific encoders through to the decision layer, combining only at the output—this preserves intra-modal structure but loses cross-modal interactions. **Intermediate fusion**, the most expressive family, stacks the two streams into a shared sequence and allows self-attention to operate across them, enabling the model to learn which acoustic moments correspond to which lexical events.

[Figure: Early, late, and intermediate multimodal fusion strategies]

Figure 5: Schematic comparison of the three main fusion philosophies for multimodal clinical speech analysis. Intermediate fusion allows cross-modal attention but requires careful temporal alignment [16].

An important refinement within intermediate fusion is explicit *cross-modal temporal alignment*. When silences precede lexical onsets, or when a patient produces a lengthened, effortful syllable spanning many acoustic frames, a naive concatenation of embeddings misaligns the two streams. The CogniAlign architecture addresses this directly: it first aligns acoustic frames to word boundaries using forced alignment, then refines the joint representation with gated cross-attention [38]. The intuition is that the network should attend to co-occurring acoustic evidence—jitter, spectral tilt shifts, articulatory closures—and lexical hypotheses together, rather than in parallel isolation.

A complementary line of work repurposes fusion architectures from adjacent paralinguistic domains—speaker verification and speech emotion recognition—and adapts them to clinical speech with domain regularisation [39]. Competitive results suggest that encoder stacks trained on large emotion corpora contribute useful priors when fine-tuned on clinical data [5].

2.3.4 Handcrafted descriptors alongside learned embeddings

Self-supervised waveform encoders are powerful but opaque. When a WavLM embedding drives a classification decision, the responsible acoustic property is hidden inside hundreds of non-linear transformations. For clinical deployment this is a genuine concern: a clinician who cannot interpret a model's reasoning cannot calibrate their trust in it.

Handcrafted descriptors address this directly. Pause histograms, intensity envelopes, fundamental-frequency variation, shimmer and jitter, harmonics-to-noise ratio, spectral moments, formant tracks, and mel-frequency cepstral coefficients all map to measurable articulatory and phonatory parameters with established clinical interpretations. Pause-oriented statistics emerge repeatedly as among the most sensitive markers of cognitive decline [40, 41]. Voice-quality perturbations (jitter, shimmer) may reflect neurogenic motor control deficits independent of the lexical content of the utterance [11].

Recent open-source pipelines demonstrate that coupling classical descriptors with modern neural backbones can preserve or improve benchmark accuracy relative to embeddings alone [28]. Whether handcrafted features provide complementary signal or merely recapitulate information already captured in the embedding depends on architecture and task: in systems with explicit audio–text alignment, part of the prosodic signal may already be encoded by the alignment mechanism, reducing the marginal contribution of auxiliary feature vectors [38, 5].

[Figure: Taxonomy of handcrafted acoustic descriptors and their clinical interpretation]

Figure 6: Handcrafted acoustic and paralinguistic descriptors grouped by feature family. Interpretability is their primary advantage over learned embeddings.

2.4 Synthesis and Positioning

Reviewing this landscape, three design commitments emerge as both well-motivated by the literature and directly addressed by the methodology of this thesis [5].

First, models should be evaluated against *pathology-aware* labels where they exist—amyloid status, variant diagnosis—rather than the simpler dementia-versus-control proxy that dominates public benchmarks. The clinical questions that matter are more specific than binary screening.

Second, competing multimodal fusion architectures should be compared under *harmonised preprocessing* on both an English reference corpus (ADReSSo) and a clinically curated Spanish-language subset aligned with SPIN. English-only results generalise poorly, and the field needs systematic cross-lingual evidence [16, 31].

Third, the complementarity of handcrafted acoustic descriptors relative to state-of-the-art self-supervised encoders and explicit audio–text alignment should be *measured empirically*, not assumed. The answer will vary by task and architecture, and quantifying it informs both model design and clinical interpretability [38, 28].

The following methodology chapter translates these three commitments into concrete architectures, training protocols, and evaluation choices.

3 Methodology

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Appendices

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